

STAT 3003: Survey Methods **- Introduction**

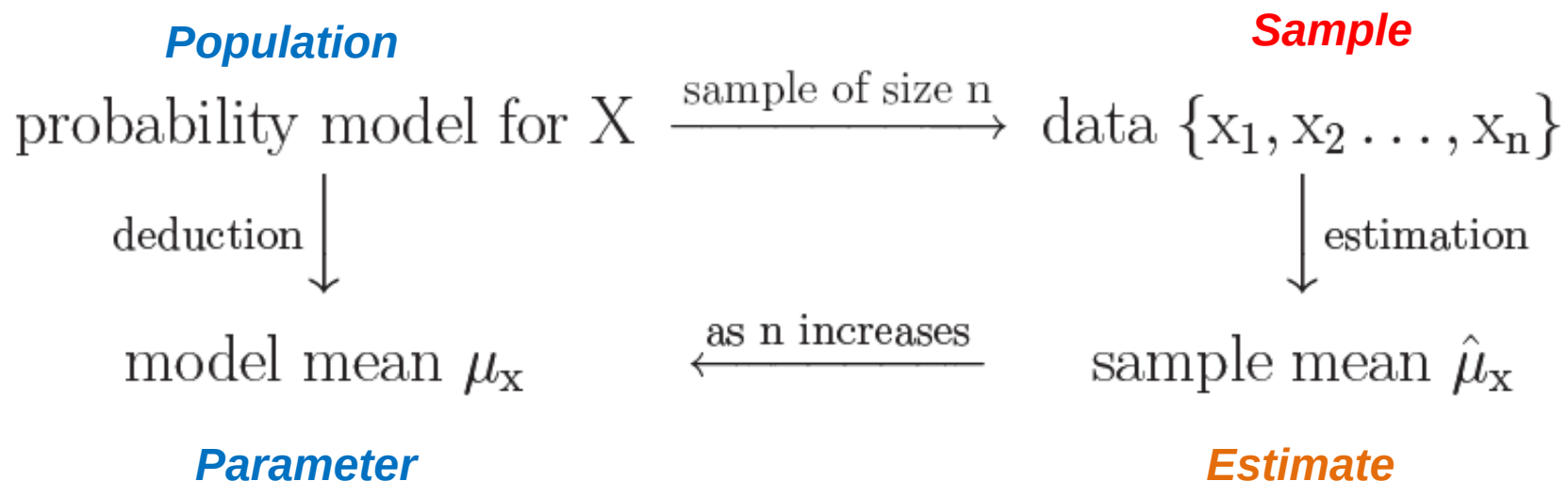
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What is Survey

- A statistical survey is any structured inquiry designed to obtain data, which may be qualitative or quantitative.
- The main goal of a survey is to collect data that is representative of the population, allowing researchers to make informed decisions or draw conclusions.
- Survey sampling describes the process of selecting a sample of elements from a target population to conduct a survey.
- Survey Methods teach how to sample elements from the population, how to construct questionnaire, how to improve the number and accuracy of responses to surveys, and how to analyse the resulting data.

Role in Statistics

- A **variable** is a characteristic measured on individuals/elements drawn from a population under study.
- **Data** are measurements of one or more variables made on a collection of individuals.
- Variable/Data Type: discrete, continuous, categorical



Role in Common Path of Statistical Studies

- 1. Scientific Goal

e.g. find disease gene, compare efficacy of 2 drugs, ...

- 2. Study Design

e.g. observation, experiment, cohort study, case-control, ...

- 3. Data Collection

e.g. what variables including confounders to collect, how to collect, ...

- 4. Data Describing

e.g. histograms, box plots, average, ...

- 5. Statistical Inference

e.g. estimation, hypothesis testing, prediction, ...

- 6. Interpretation

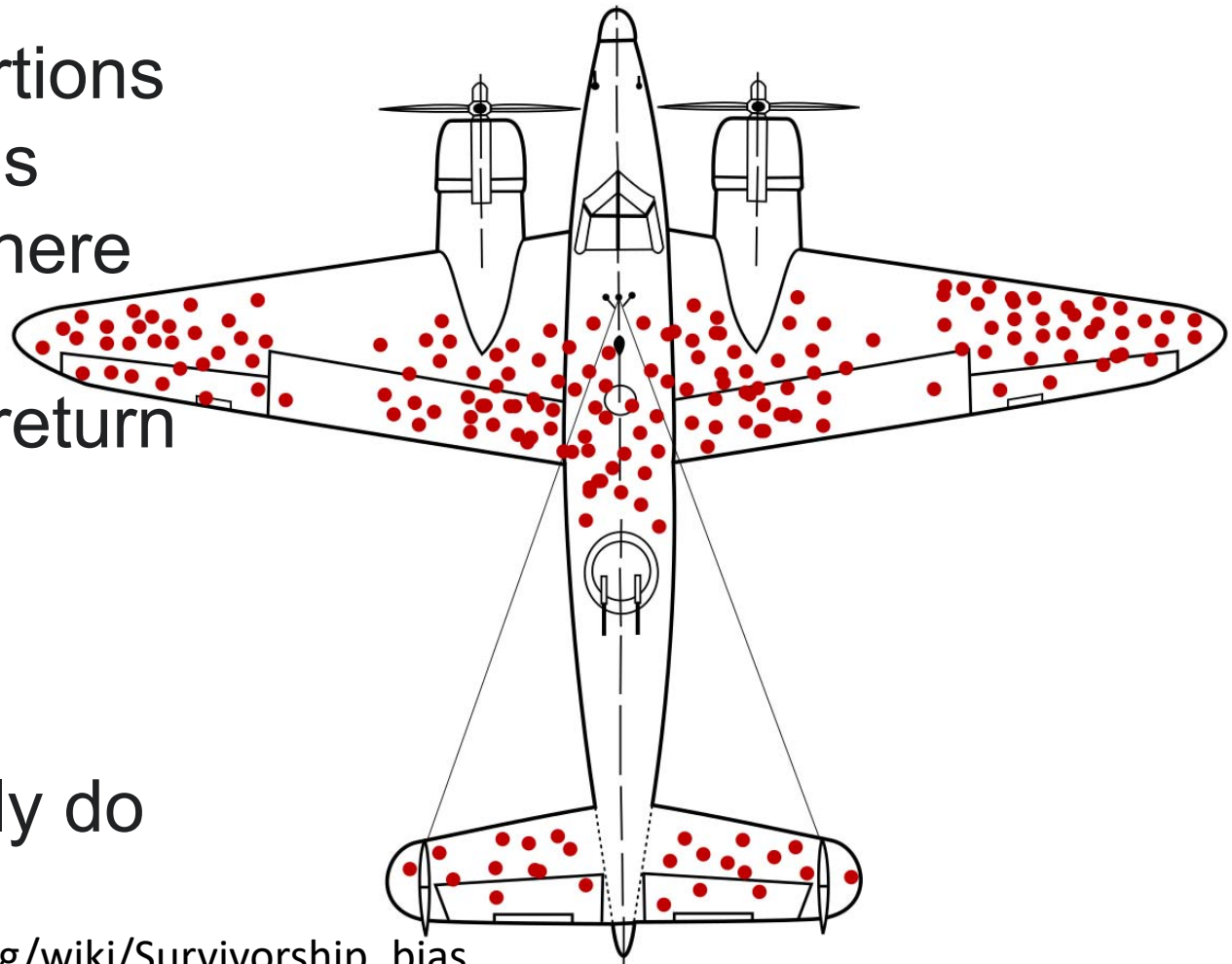
Motivating Example 1:

IMPORTANT to **understand your data:**

Survival Bias - a form of selection bias

The damaged portions
of returning planes
show locations where
they can sustain
damage and still return
home.

Those hit in other
places presumably do
not survive.



Ref: https://en.wikipedia.org/wiki/Survivorship_bias

Motivating Example 2:

IMPORTANT to **understand your data:**
selection bias and nonresponse

Example 1.1 from Lohr book: **“The Survey That Killed a Magazine”** :

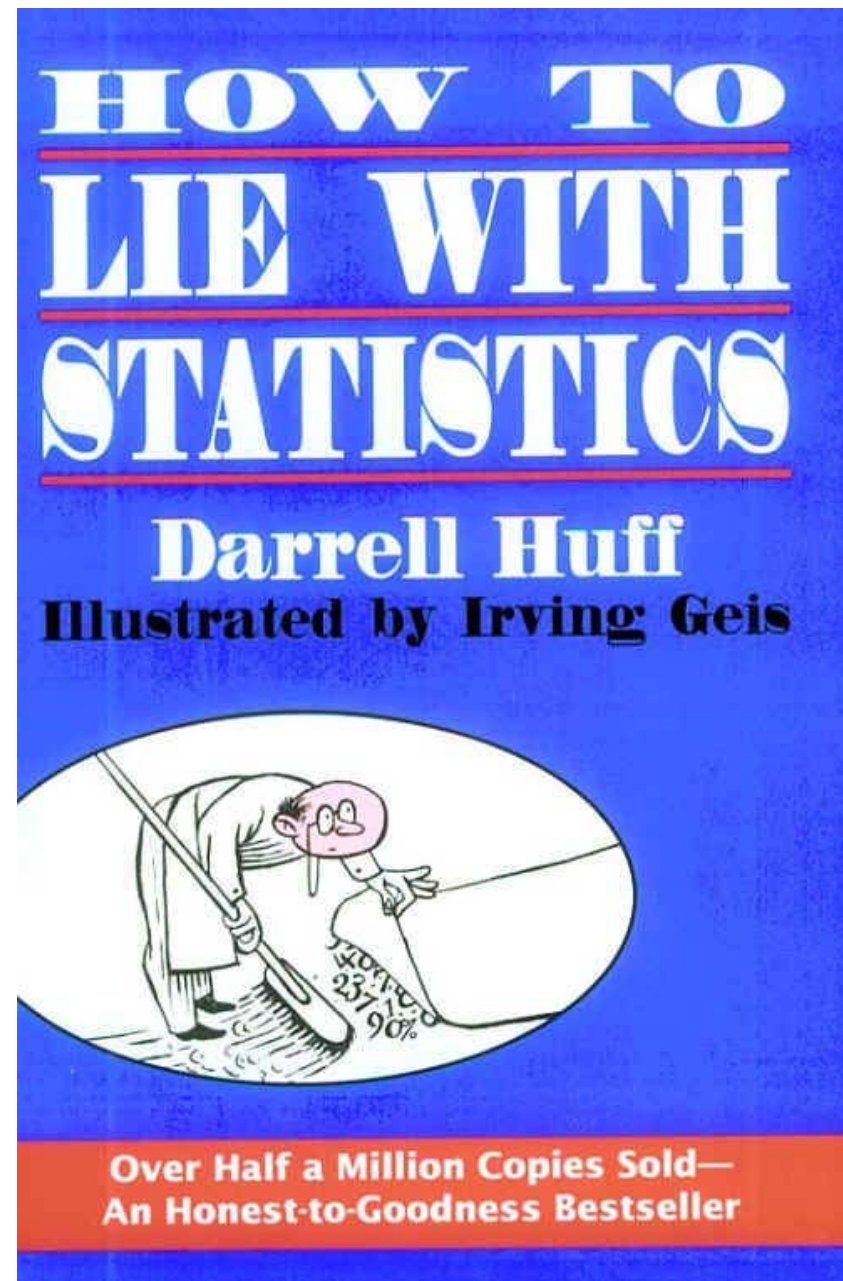
The Literary Digest, a weekly news magazine founded in 1890, did a presidential election poll in 1936 and predicted that Republican Alf Landon would win, but the later election showed that Democrat Franklin Roosevelt won with a big margin.

*names of persons to be polled were compiled from sources such as telephone directories and automobile registration lists => **bias** to rich people

* Ten million questionnaires were mailed out, and more than 2.3 million were returned => **nonresponse**, and persons supporting Landon were much **more likely to response** to the survey

Quotations

- There are three kinds of lies: lies, damn lies, and statistics.
--Mark Twain => Benjamin Disraeli
- It is easy to lie with statistics, but easier to lie without them.
--Frederick Mosteller



alert for statistics

- When statistics are not based on strictly accurate calculations, they mislead instead of guide. The mind easily lets itself be taken in by the **false appearance of exactitude** which statistics retain in their mistakes, and confidently adopts **errors clothed in the form of mathematical truth**.

—Alexis de Tocqueville, *Democracy in America*

Syllabus/Course Outline

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Logistics

- Homework: announce in class, available online, due before class, **submit through Blackboard**, R is recommended for programming
- Midterm policy: closed book, but allow 1 page note (A-4, double side)
- Final Exam policy: close-book, but allow 1 page note (A-4, double side)

Reminder:

- This course assumes basic statistics and probability background, and you will have to deal with quite many formulas.
- Life is short, so use it on useful and/or interesting things
- If you select it, devote to it. Don't waste your time
- Respect others' right to learn, as well as respect yourself

Background Checking

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Introduction Contents

- 1.1 The Problem
 - The goal of sampling, potential errors, example
- 1.2 Definitions of Population Parameters
 - μ, σ^2, S^2 (the corrected population variance),
 $Cov(X, Y), \rho$
- 1.3 Estimation
 - $\theta, \hat{\theta}$, sampling distribution, error of estimation,
margin of error, n , confidence interval (CI), bias, MSE
- 1.4 Survey Terminology
 - Element, population, sampling unit, frame, sample,
example

1.1 The Problem

- How to infer facts about a population from the information within a sample?
 - The average height of statistics students in Hong Kong
 - The proportion of statistics students in Hong Kong who wear glasses
 - The number of statistics students in Hong Kong who support the use of ChatGPT for homework

1.1 The Problem

- How to infer facts about a population from the **information** within a sample, *given a budget?*
- Each observation (a measurement, answers on a questionnaire) contains information
- But each observation also **costs** something (time and money)

1.1 The Problem

- Our resources are finite
- Too little information and we will inevitably make a poor estimate
- Too much information means we have wasted time/money
- What does the amount of information in a sample depend on?

1.1 The Problem

- The amount of information in a sample depends on the **sample size** and the **amount of variation** in the data
- In this course we will see many methods to control the amount of variation – these are called **sample survey designs**

1.1 The Problem

- Survey design and sample size determine the amount of information in the sample, assuming that each element in the sample is measured accurately
 - How much do you earn?
 - “Does it seem possible to you that the Holocaust never happened?” versus “Does it seem possible or does it seem impossible to you that the Nazi extermination of the Jews never happened?”
 - Have you ever copied your homework?

1.1 The Problem

- How much do you earn?
 - People tend to inflate their true wage to impress the person conducting the questionnaire
- Does it seem possible or does it seem impossible to you that the Nazi extermination of the Jews never happened?
 - The double negative is confusing. A clearer version of this question reduced the 'Yes' response from 22% to 1%
- Have you ever copied your homework?
 - This question is deeply embarrassing to answer

1.1 The Problem

- Problems like these can't be handled using statistical techniques which process the data after it has been collected
- To address these kinds of errors, the questionnaire/survey should be **planned carefully**
- We shall see methods to do this

1.1 The Problem

- We will want to **summarize information** about populations and samples
- We treat the results obtained in sample surveys as **random**, because we will **randomly select** members of the population to be in our sample
 - Why would we want to do this?
- Therefore different people using the same methods can obtain different results

1.1 The Problem

- Example – The McDonalds Survey
 - Our survey consists of one question: “How many times a month do you eat food from McDonalds?”
 - The population is everyone in this lecture theatre, now
 - Answer the question and write down your answer

1.1 The Problem

- Example – The McDonalds Survey
 - Our survey consists of one question: “How many times a month do you eat food from McDonalds?”
 - The population is everyone in this lecture theatre, now
 - Answer the question and write down your answer
 - I need 3 volunteers...

1.1 The Problem

- Example – The McDonalds Survey
 - Each volunteer is interested in how much McDonalds food this class consumes
 - But they do not have enough time or much to ask everyone
 - So they randomly select a sample of 3 each
 - And they collect their results...

1.1 The Problem

- Example – The McDonalds Survey
 - What can they do with their results?
 - With the information they have, how can they summarize their sample?
 - Can they infer anything about the population from their data?

1.1 The Problem

- The McDonalds Survey
 - They might average the responses they received
 - They might work out the standard deviation of the responses they received
 - Then each volunteer will have **estimates** of the population mean and standard deviation
 - Let's compare their estimates to these population **parameters**

1.1 The Problem

- Example – McDonalds Survey
 - Think about: how could the volunteers improve their surveys?

1.2 Definitions of Population Parameters

- We introduced population mean and population standard deviation and their sample versions, the sample mean and sample standard deviation
- Before we start creating survey designs, we need to remember some fundamental definitions and concepts from statistics

1.2 Definitions of Population Parameters

- **Population Mean**

- Let there be N people in the population. Let the variable of interest be Y (e.g. height in cm) and let person i have value u_i (e.g. the 1729th person has height 180 cm)
- Is the **average value** of all measurements in the population, i.e. $\frac{u_1 + u_2 + \dots + u_N}{N} := \mu$
 - We will often denote the population mean by μ
- The population mean in our McDonalds example is...

1.2 Definitions of Population Parameters

- **Finite Population Variance**

- The finite population variance is given by

- $$\frac{1}{N} \sum_{i=1}^N (u_i - \mu)^2 := \sigma^2$$

- We often denote the population variance by σ^2

- The **population standard deviation** (SD) simply the square root of the population variance

- $$\sqrt{\sum_{i=1}^N \frac{(u_i - \mu)^2}{N}} := \sigma$$

- The population standard deviation in our McDonalds example is...?

1.2 Definitions of Population Parameters

- In some textbooks, we may see finite population variance defined as

$\frac{1}{N-1} \sum_{i=1}^N (u_i - \mu)^2 := S^2$, the so-called “corrected population variance”

- Comparing this definition to our σ^2 , notice

$S^2 = \left(\frac{N}{N-1}\right) \sigma^2$, so as N become large, they are more or less the same

- Why bother with S^2 ?

Because the “ $\frac{1}{N-1}$ ” factor simplifies many formulas, as we’ll see later.

- I prefer σ^2 to S^2 as it better reflects the idea of variance as an expectation

$$\text{Var}(Y) = E[(Y - \mu)^2]$$

1.2 Definitions of Population Parameters

- It will be useful to have some measure of dependence between two random variables
- Some variables are clearly **independent**
 - Level of the Ulan Bator Stock Index and my shoe size
- But some obviously have some **dependence**
 - FTSE100 stock index and UK gilt prices
- **Covariance** and **Correlation** help us quantify dependence

1.2 Definitions of Population Parameters

- **Covariance**

- We have a finite population of size N . Each person in the population has two variables of interest, Y and X
- Let the population means of Y and X are μ_Y and μ_X respectively. Then the population covariance of X and Y is given by
 - $\frac{1}{N} \sum_{i=1}^N [(y_i - \mu_Y)(x_i - \mu_X)] := \text{Cov}(X, Y)$
- If X and Y are independent, then $\text{Cov}(X, Y)$ is zero
- The converse is not necessarily true
- It is tricky to decide whether a covariance is large or small because the scales of y and x are combined
- It would be better to have a dimensionless number to describe dependence...

1.2 Definitions of Population Parameters

- **Correlation**
 - Is that dimensionless quantity
 - The correlation between X and Y is just the covariance of X and Y divided by the standard deviations of X and Y
 - That is, $\frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y} := \rho$
 - This must lie between -1 and 1 and it has no units
 - We often denote it by ρ (rho)

1.2 Definitions of Population Parameters

- Population mean, variance and correlation are examples of parameters
 - **Parameters** are numerical descriptive measures of a population (mean, **total**, variance, **proportion**, **population size**, etc.)
 - Unless we have measured every member of the population, parameters will always be **unknown**

1.3 Estimation

- Much of this course will be spent on investigating how we can best **estimate** the parameters of a population from a sample of the population
 - We'll take the observations from our sample, plug them into a formula that gives our guess for the value of the parameter
 - That guess from our sample is called the **estimate**. The general formula is called the **estimator**.

1.3 Estimation

- Example
 - From our McDonald's surveys, what were the **parameters**, **estimates** and **estimators**?
- We denote a generic parameter by θ (theta)
- We denote an estimator of that parameter by putting a hat on it - $\hat{\theta}$
- Notice that, before we conduct the survey, $\hat{\theta}$ is a **random variable**
 - Because the elements in the sample were chosen **randomly**
- Thus $\hat{\theta}$ will have a probability distribution
 - This distribution is called the **sampling distribution**

1.3 Estimation

- Why do we care about the sampling distribution?
 - Because it is not enough to just present a single number as your best guess for the value of θ
 - This gives no idea about how good or accurate your guess is
- Your estimate for θ is very unlikely to be exactly correct and indeed, we wouldn't know if it was, because we'll never know the true value of θ
 - But it would be nice to know that the chances of it being wildly incorrect are quite small

1.3 Estimation

- That is, we would like our guess to be close to θ , most of the time
 - i.e. we would like $|\hat{\theta} - \theta|$ to be smaller than, say, some number d , most of the time
 - $|\hat{\theta} - \theta|$ is called the **error of estimation**
- We can write this in terms of a probability statement
 - We would like $\Pr(|\hat{\theta} - \theta| < d) = 1 - \alpha$, where α is a small number like 0.05 or 0.01 and d is called the **margin of error**
- The only random variable in the statement above is $\hat{\theta}$. In order to choose a sensible number d , we must have some idea **what the sampling distribution is**

1.3 Estimation

- The distribution of $\hat{\theta}$ will depend crucially on n , the size of the sample
- If n is small, the distribution of $\hat{\theta}$ is usually complicated and unknown
- If n is large, the distribution of $\hat{\theta}$ is much more regular and is well approximated by distributions we know well
- If n is large and close to N , the distribution of $\hat{\theta}$ tends to concentrate more and more around the true value of θ , but is not well approximated by distributions we know well

1.3 Estimation

- Imagine we are confident enough to say $\frac{\hat{\theta} - E[\hat{\theta}]}{\sigma_{\hat{\theta}}} \sim N(0,1)$, where $\sigma_{\hat{\theta}}$ is the standard deviation of $\hat{\theta}$
- Then, we may write

$$P\left(\frac{\hat{\theta} - E[\hat{\theta}]}{\sigma_{\hat{\theta}}} < z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$
$$\Rightarrow P\left(-z_{1-\frac{\alpha}{2}} < \frac{\hat{\theta} - E[\hat{\theta}]}{\sigma_{\hat{\theta}}} < z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

1.3 Estimation

- Carrying on, we have

$$\Rightarrow P\left(-\sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < \hat{\theta} - E[\hat{\theta}] < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(-\hat{\theta} - \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < -E[\hat{\theta}] < -\hat{\theta} + \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\hat{\theta} - \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < E[\hat{\theta}] < \hat{\theta} + \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

- And if $\hat{\theta}$ is unbiased, i.e. if $E[\hat{\theta}] = \theta$, then

$$P\left(-\sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}} < \hat{\theta} - \theta < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(|\hat{\theta} - \theta| < \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right) = 1 - \alpha, \text{ meaning we have found}$$

our sensible number, d

1.3 Estimation

- The interval $\left(\hat{\theta} - \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}, \hat{\theta} + \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}\right)$ is random
 - recall $\hat{\theta}$ is a random variable
 - If we plug in our observation of $\hat{\theta}$ from our sample, we have a **$100 \times (1 - \alpha)\%$ confidence interval for parameter θ**
- Having a **normal sampling distribution** is very handy. We will discuss conditions for its existence in the next lecture.
- Clearly we want our estimator $\hat{\theta}$ to have the property “ $E[\hat{\theta}] = \theta$ ”. What else do we want from our estimator?

1.3 Estimation

- We want our observation of $\hat{\theta}$ from our sample to be close to the parameter we're interested in
- This can happen if
 - The estimator generates estimates **are correct on average**
 - And these estimates are **not too spread out**
- Statistically, we want
$$E[\hat{\theta}] = \theta, \text{ as before}$$
$$E[(\hat{\theta} - \theta)^2] \text{ is small}$$

1.3 Estimation

- The quantity $E[\hat{\theta}] - \theta$ is called the **bias** of the estimator $\hat{\theta}$
 - We want our estimators to be **unbiased**, i.e. to have zero bias
- The quantity $E[(\hat{\theta} - \theta)^2]$ is called the **mean square error** (MSE) of $\hat{\theta}$
- There is a relationship between MSE, bias and variance

1.3 Estimation

- $$\begin{aligned} E \left[(\hat{\theta} - \theta)^2 \right] &= E \left[(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta)^2 \right] \\ &= E \left[(\hat{\theta} - E[\hat{\theta}])^2 \right] + 2(E[\hat{\theta}] - \theta)E \left[\hat{\theta} - E[\hat{\theta}] \right] \\ &\quad + (E[\hat{\theta}] - \theta)^2 \\ &\Rightarrow \textcolor{red}{MSE}(\hat{\theta}) = \textcolor{red}{Var}(\hat{\theta}) + \textcolor{red}{Bias}(\hat{\theta})^2 \end{aligned}$$
- Therefore, **if estimator $\hat{\theta}$ is unbiased** (most of our estimators will be unbiased), then its **MSE equals its variance**

1.3 Estimation

- Here are three useful words to describe an estimator
 - Unbiased if $E[\hat{\theta}] = \theta$
 - **Precise** if $Var(\hat{\theta})$ is small
 - **Accurate** if $E[\hat{\theta}] = \theta$ and $E[(\hat{\theta} - \theta)^2]$ is small
- Thus a precise estimator may not be accurate because it is biased.
- Precision considers how close estimates from different samples are to each other.
- Accuracy considers how close estimates are to the true value.

1.4 Survey Terminology

- We will go through some unavoidable technical terms which we will use repeatedly throughout the course
- **Element**
 - An element is an object on which a measurement is taken
 - E.g. a Hong Kong statistics student, an orange, anything

1.4 Survey Terminology

- **Population**
 - A collection of elements about which we want to make inferences
 - E.g. Hong Kong statistics students, oranges grown in California

1.4 Survey Terminology

- **Sampling Units**

- Non-overlapping collections of elements from the population that cover the entire population (i.e. they partition the population)
 - E.g. CUHK statistics students, HKU stats students, HKUST stats students... etc. (assuming that a student cannot study statistics in two university in Hong Kong at the same time)
 - Often sampling units and elements are the same

1.4 Survey Terminology

- **Frame**
 - A frame is a list of sampling units
 - E.g. Lists of registered CUHK statistics students, of HKU statistics students, etc.
 - ‘List’ can be loosely interpreted: any representation of the population that allows random selection of units can be a frame

1.4 Survey Terminology

- Sample
 - A collection of sampling units drawn from a single or from multiple frames

1.4 Survey Terminology

- Example: I want to infer the proportion of Hong Kong high school students who support the use of ChatGPT for homework
- I decide I will ask 200 students
- I have a list of all the high schools in HK. I select 25 of these
- For each of the 25, I collect a name list from the school and I select 8 to interview
- What are the **element**, **population**, **sampling units**, **frame(s)** and **sample** in this example?

STAT3003: Survey Methods

2. Simple Random Sampling

Contents

- 2.1 Probability vs Quota
 - Quota sampling, 1948 US election, lessons
- 2.2 Simple Random Sampling
 - Definition, examples, inclusion probability π_i , methods
- 2.3 Properties of $\bar{Y}$
 - Definition, $E(\bar{Y})$, $Var(\bar{Y})$, sample variance $\hat{\sigma}^2$, observed sample variance s^2 , $\widehat{Var}(\bar{Y})$
- 2.4 Confidence Intervals for μ
 - Central Limit Theorem, t -distribution, normality, example

Contents

- 2.5 Sample sizes for μ
 - Using a normal percentile, choosing a value for σ , using a t -percentile via Excel/R etc
- 2.6 Estimating τ
 - $\hat{\tau}$, bias, $Var(\hat{\tau})$, $\widehat{Var}(\hat{\tau})$, CIs, sample size, example
- 2.7 Estimating p
 - $\hat{p}$, $Var(\hat{p})$, $\widehat{Var}(\hat{p})$, CIs, sample size, example
- 2.8 Using SRS in Practice
 - Simplest but not the best

2.1 Probability vs Quota

- In the previous lecture, we said our sample would consist of elements **randomly chosen** from the population
- Samples based on planned randomness are called **probability samples**
- We saw that probability samples allowed us to establish estimator bias/unbiasedness and make **probability statements** about our estimators
 - E.g. confidence intervals

2.1 Probability vs Quota

- Why use probability samples? Why not just choose elements carefully to ensure a **representative** sample?
- Samples which select elements to ensure the sample has **very similar characteristics** as the population are called **quota samples**
 - E.g. select a sample which has the same proportions of sex, age, income, education etc. as the population
- Quota sampling was commonly used by **political pollsters** in the US up to and including the 1948 election

2.1 Probability vs Quota

Characteristic	Population Estimate (%)	Gallup Sample (%)	Roper Sample (%)
Sex – Male	49.1	50.5	50.1
Sex – Female	50.9	49.5	49.9
Age – 21-34	34.3	29.6	36.3
Age – 35-49	30.9	34.4	30.8
Age – 50+	34.8	35.9	32.9
Colour – White	96.1	95.0	97.4
Colour – Black	3.9	5.0	2.6
Etc.	⋮	⋮	⋮
# Respondents		2972	3501

Extract of Table 2.2 from Scheaffer et al. 2012: Comparison of Gallup and Roper samples with U.S. population

2.1 Probability vs Quota

- The 1948 presidential election was between Thomas Dewey and Harry Truman
- The Gallup and Roper polls used **quota sampling** to find samples similar to the overall population in terms of
 - Sex, age, education, colour, veteran status, labour union membership
- The polls predicted a comfortable victory for Dewey
 - Gallup had Dewey 49.5% to Truman's 44.5%
 - Roper had Dewey 52.2% to Truman's 37.1%

2.1 Probability vs Quota

- The result was Dewey 45.1% to Truman 49.5%
 - It seems the quota sampling dramatically overrepresented Dewey supporters (Republicans)
- How?
 - Instructions to interviewers in a quota sampling poll tended to be like “Find 2 men and 3 women in your block. Make sure 4 are over 25 years old and one is under 25.”
 - Instructions like these leave lots of room for human **subjectivity**. We want **objectivity**.
 - Perhaps the interviewer subconsciously (or consciously!) seeks out people who look “nicer” – e.g. richer and if Republicans tend to have higher incomes then...

2.1 Probability vs Quota

- Maybe the quota sampling wasn't specific enough
 - Why not include more characteristics, like income level, marital status, attitude towards issues like taxes, healthcare, foreign policy, etc.
 - Or combinations of the above, e.g. match the ratio of white, single males with union membership
- Clearly it is impossible to control all factors which may influence a person's preference of Dewey or Truman
 - Indeed, trying to fix one ratio might disturb other ratios
- Randomisation balances out all factors inherent in a population
 - including the factors you know but would like to control
 - those you know but can't control
 - and those you don't know you don't know

2.1 Probability vs Quota

- In this course, **all sample designs will be based on probability sampling**
 - All modern samples use probability sampling
- They produce more accurate results than quota sampling designs
- Also, they
 - Require a smaller sample size
 - Are easier to do
 - Are much more objective

2.1 Probability vs Quota

- What do you think of these sampling methods? (Exercise 4.5 in Scheaffer et al.)
 - A famous newspaper columnist estimates the proportion of unhappy marriages by asking her readers to write to her.
 - To find the average length of string in a bag, a student reaches in, mixes up the strings, selects one, mixes them up again, selects another, and so on

2.2 Simple Random Sampling

- This is our first **sample survey design**, i.e. a procedure for selecting a sample
- Let the population have size N
- **Simple Random Sampling** (SRS) is
 - Random sampling **without replacement** such that every possible sample of n units is **equally likely**
- Therefore, there are $\frac{N!}{n!(N-n)!}$ possible random samples and each occurs with the same probability

2.2 Simple Random Sampling

- Decide if the following sampling methods produce a SRS (Exercise 4.11 from Scheaffer et al.)
 - A class has 15 boys and 15 girls. Assign the boys numbers from 1 to 15, the girls from 16 to 30. Use a random number table to select six numbers from 1 to 30. Select the students with those numbers
 - Same class as above. This time use the random number table to select 3 numbers from 1 to 15 and 3 from 16 to 30
 - In a full class with six rows of five chairs each, randomly choose a row and select all students in that row

2.2 Simple Random Sampling

- Given a population of size N , what is the probability the i th element is in a SRS of size n ?
 - This is called the **inclusion probability** of element i
 - Let $\pi_i = P(\text{Element } i \text{ is in the sample})$
 - Then write **$\pi_i = E[Z_i]$** where Z_i is a random variable which equals 1 if element i is in the sample or 0 otherwise
 - $= \sum_{\text{All samples containing element } i} P(\text{Sample occurs})$
 - $= \sum_{\text{All samples containing element } i} \frac{1}{\frac{N!}{n!(N-n)!}}$

2.2 Simple Random Sampling

- Now say element i is in sample of size n
 - We have to fill the remaining $(n - 1)$ spaces from $(N - 1)$ elements in the population
 - There are $\left(\frac{(N-1)!}{(n-1)!(N-n)!}\right)$ ways to do this
 - Hence there are $\frac{(N-1)!}{(n-1)!(N-n)!}$ samples with element i
- Therefore, we have

$$- \pi_i = \frac{(N-1)!}{(n-1)!(N-n)!} \times \frac{1}{\frac{N!}{n!(N-n)!}} = \frac{n}{N} = E[Z_i]$$

2.2 Simple Random Sampling

- How to perform simple random sampling?
- Don't
 - Use your own judgement to “randomly” select (so-called **haphazard sampling**)
 - Try to select a “representative” sample
- Because
 - Both are prone to **investigator bias**
 - Both lead to estimators whose properties we don't know.

2.2 Simple Random Sampling

- How to perform simple random sampling?
- Do
 - Draw lots
 - Use random numbers in a book or Excel/R...
- Example
 - I want to draw a **SRS without replacement** of 10 students from a population of 21 students

2.2 Simple Random Sampling

- 1st Method: Draw Lots
 - Prepare 21 cards
 - Write a student number on each
 - Shuffle well
 - Pick up 10 cards

2.2 Simple Random Sampling

- 2nd Method: Use a [Random Numbers Table](#)
 - Find a page of random numbers
 - Assign the students numbers from 01 to 21
 - Randomly select a starting point in the random number table (drop a pencil, see where the pointed end points to)
 - Read off pairs of numbers in any direction you like
 - Sample without replacement

2.2 Simple Random Sampling

- 3rd Method: Use **Random Numbers in Excel/R...**
 - Can be done equivalently with other spreadsheet programs
 - Create a column of the 21 students ID numbers or names
 - Alongside, create a column of 21 **random numbers** using =RAND()
 - Use **Sort** to arrange the students names in order of ascending random numbers
 - Choose the first 10 names

2.2 Simple Random Sampling

- We've seen how to conduct simple random sampling
- What parameters might we be interested in?
We will consider three
 - Population mean μ
 - Population total τ
 - Population proportion p

2.2 Simple Random Sampling

- We shall see that τ and p can be written easily in terms of μ
- So we shall work out the following for μ and transfer the results to τ and p
 - What shall we use as an estimator $\hat{\mu}$ for μ ?
 - Is $\hat{\mu}$ unbiased?
 - What is the variance of $\hat{\mu}$?
 - How can we estimate the variance of $\hat{\mu}$?
 - How can we calculate confidence intervals for μ ?
 - How can we determine an appropriate sample size n for estimating μ ?

2.3 Properties of $\bar{Y}$

- Our estimator for μ is what we expect: **the sample mean**
- Let the population have values $\{u_1, u_2, \dots, u_N\}$
- Let $\{Y_1, Y_2, \dots, Y_n\}$ be the sample drawn by SRS without replacement
- The sample mean is given $\bar{Y} := \frac{1}{n} \sum_{i=1}^n Y_i$
 - We may write this another way: $\bar{Y} = \frac{1}{n} \sum_{j=1}^N u_j Z_j$

2.3 Properties of $\bar{Y}$

- We calculate $E(\bar{Y})$

$$\blacktriangleright = E\left(\frac{1}{n}\sum_{j=1}^N u_j Z_j\right) = \frac{1}{n}\sum_{j=1}^N u_j E(Z_j)$$

- Now, since $E(Z_j) = \frac{n}{N}$ for all j

$$\blacktriangleright E(\bar{Y}) = \frac{1}{n}\sum_{j=1}^N u_j \frac{n}{N} = \frac{1}{N}\sum_{j=1}^N u_j = \mu$$

- Thus $E(\bar{Y}) = \mu$ and our estimator is unbiased.
- This is good.

2.3 Properties of $\bar{Y}$

- Before we calculate $Var(\bar{Y})$, let us recall a useful result
 - Let $X_1, \dots, X_n$ be random variables and $a_1, \dots, a_n$ be constants
 - Then for $U := \sum_{i=1}^n a_i X_i$, we have
 - $Var(U) = \sum_{i=1}^n a_i^2 Var(X_i) + \sum_{i=1}^n \sum_{j \neq i} a_i a_j Cov(X_i, X_j)$
- Therefore, according to our useful result
 - $Var(\bar{Y}) = \sum_{j=1}^N \left(\frac{u_j}{n}\right)^2 Var(Z_j) + \sum_{j=1}^N \sum_{k \neq j} \frac{u_j}{n} \frac{u_k}{n} Cov(Z_j, Z_k)$
- Since Z_j is a Bernoulli random variable with probability of “success” $\frac{n}{N}$
 - $Var(Z_j) = \frac{n}{N} \left(1 - \frac{n}{N}\right)$

2.3 Properties of $\bar{Y}$

- Note
 - $Cov\left(Z_j, Z_k\right) = E\left(Z_j Z_k\right) - E\left(Z_j\right) E\left(Z_k\right) = E\left(Z_j Z_k\right) - \left(\frac{n}{N}\right)^2$
 - Note $Z_j Z_k$ equals 0 unless both elements j and k are in the sample, then $Z_j Z_k = 1$
 - The number of possible samples containing both j and k is $\frac{(N-2)!}{(n-2)!(N-n)!}$
 - Each SRS sample is equally likely to occur, with the number of possible random samples $\frac{N!}{n!(N-n)!}$
 - Hence $E\left(Z_j Z_k\right) = \frac{(N-2)!}{(n-2)!(N-n)!} \times \frac{1}{\frac{N!}{n!(N-n)!}} = \frac{n(n-1)}{N(N-1)}$

2.3 Properties of $\bar{Y}$

- Therefore

- $Cov(Z_j, Z_k) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)} = -\frac{n(1-\frac{n}{N})}{N(N-1)}$

- Thus

$$Var(\bar{Y}) = \frac{n}{N} \left(1 - \frac{n}{N}\right) \sum_{j=1}^N \frac{u_j^2}{n^2} - \frac{n}{N} \left(1 - \frac{n}{N}\right) \frac{1}{N-1} \sum_{j=1}^N \sum_{k \neq j}^N \frac{u_j}{n} \frac{u_k}{n}$$
$$\Rightarrow Var(\bar{Y}) = \frac{1}{n^2} \frac{n}{N} \left(1 - \frac{n}{N}\right) \left[\sum_{j=1}^N u_j^2 - \frac{1}{N-1} \sum_{j=1}^N \sum_{k \neq j}^N u_j u_k \right]$$

2.3 Properties of $\bar{Y}$

- Now use the **identity**
 - $\sum_{j=1}^N (u_j - \mu)^2 = \frac{1}{N} [(N-1) \sum_{j=1}^N u_j^2 - \sum_{j=1}^N \sum_{j \neq k} u_j u_k]$
(exercise!)
 - $\Rightarrow Var(\bar{Y}) = \frac{1}{n^2} \frac{n}{N} \left(1 - \frac{n}{N}\right) \frac{N}{N-1} \sum_{j=1}^N (u_j - \mu)^2$
 - $\Rightarrow Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$
- We have found the variance of our estimator
 - How is this different from the formulas you have seen before?

2.3 Properties of $\bar{Y}$

- We have found the variance of our estimator
- Notice the difference between this and the “usual” variance of the sample mean – there is a factor of $\left(\frac{N-n}{N-1}\right) = 1 - \frac{n-1}{N-1}$
- This is an example of a finite population correction, or a **f.p.c.**
 - We will also see the f.p.c. $\frac{N-n}{N} = 1 - \frac{n}{N}$ appear more commonly in our equations, which uses the **sampling fraction** $f = \frac{n}{N}$
- Thus a sample of 15 out of a population of 25 contains more information (less variability) than a sample of 15 out of a population of 25,000

2.3 Properties of $\bar{Y}$

- Unfortunately, most of the time **we do not know parameter σ**
- Can we estimate this as well?
- Let us use the **sample variance**
 - $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$
 - Denote by s^2 the **observed value** of $\hat{\sigma}^2$, i.e. $s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$
- Is this an unbiased estimator for σ^2 ?

2.3 Properties of $\bar{Y}$

- By repeating a similar calculation to showing $Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$, we can show
 - $E[\hat{\sigma}^2] = \frac{N}{N-1} \sigma^2 = \frac{1}{1-\frac{1}{N}} \sigma^2$
 - (See Problem Sheet 1)
- Hence, $\hat{\sigma}^2$ is a biased estimator of σ^2
 - But $\left(1 - \frac{1}{N}\right) \hat{\sigma}^2$ is an unbiased estimator for σ^2

2.3 Properties of $\bar{Y}$

- Since $Var(\bar{Y}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)$, we see that
 - $\frac{1}{n} \frac{N-n}{N-1} \frac{N-1}{N} \hat{\sigma}^2 = \left(1 - \frac{n}{N} \right) \frac{1}{n} \hat{\sigma}^2$ is an unbiased estimator for $Var(\bar{Y})$
- As this quantity will be used rather often, we will give it its own notation:
 - $\widehat{Var}(\bar{Y}) = \left(1 - \frac{n}{N} \right) \frac{\hat{\sigma}^2}{n}$
 - Notice the appearance of the f.p.c.
- What happens as the population size grows large?

2.4 Confidence Intervals for μ

- Let's recap

- Our estimator $\hat{\mu}$ for parameter μ is the sample mean, $\bar{Y}$.
- We showed $\bar{Y}$ is unbiased. This means its Mean Square Error is equal to its variance.
- We found the variance of $\bar{Y}$. It depends on the population variance σ^2 .
- Since we usually do not know the true value of σ^2 , we will usually not know the true variance of $\bar{Y}$.
- So we found an unbiased estimator for the variance of $\hat{\mu}$.

2.4 Confidence Intervals for μ

- Knowing about the unbiasedness and variance of $\bar{Y}$ is indeed useful.
- But to create confidence intervals, we need to know about its **sampling distribution**
- Recall what we said before in “Introduction”. If the parameter is θ and the estimator is $\hat{\theta}$, it would be nice if
 - $\hat{\theta}$ is unbiased and
 - $\frac{(\hat{\theta} - E[\hat{\theta}])}{\sigma_{\hat{\theta}}} \sim N(0,1)$

2.4 Confidence Intervals for μ

- If $\hat{\theta}$ has these properties, then we can easily find a $100(1 - \alpha)\%$ confidence interval for θ
 - $\hat{\theta} \pm \sigma_{\hat{\theta}} z_{1-\frac{\alpha}{2}}$
- Let θ be μ and $\hat{\theta}$ be $\bar{Y}$
 - We have already shown $\bar{Y}$ is unbiased.
 - Can we say $\frac{\bar{Y} - \mu}{\sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}} \sim N(0,1)$?

2.4 Confidence Intervals for μ

- Yes, sometimes, according to the central limit theorem (CLT) for simple random sampling without replacement from finite populations Hájek (1960)
 - If certain technical conditions hold (!) and if n , N and $N - n$ are all sufficiently large,
 - then $\frac{\bar{Y} - \mu}{\sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}}$ is approximately $N(0,1)$ and
 - $\bar{Y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N-1} \frac{\sigma^2}{n}}$ is a large-sample $100(1 - \alpha)\%$ C.I.

2.4 Confidence Intervals for μ

- But usually we don't know the value of σ
 - Since $\frac{N-1}{N} \hat{\sigma}^2$ is an unbiased estimator for σ^2 , it is common to **replace σ** in the formula with $\sqrt{\frac{N-1}{N}} \hat{\sigma}$
 - And the C.I. becomes $\bar{Y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N}} \frac{\hat{\sigma}}{\sqrt{n}}$.
 - Note this is a random variable. The observed value is $\bar{y} \pm z_{1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N}} \frac{s}{\sqrt{n}}$.

2.4 Confidence Intervals for μ

- But hang on, that assumes
 - $\frac{\bar{Y}-\mu}{\sqrt{\frac{N-n}{N}\left(\frac{\hat{\sigma}^2}{n}\right)}} = \frac{\bar{Y}-\mu}{\sqrt{\widehat{Var}(\bar{Y})}} \sim N(0,1)$, which isn't true...
- However, we do know that if the $Y_1, Y_2, \dots, Y_n$ are i.i.d. normal, then
 - $\frac{\bar{Y}-\mu}{\sqrt{\widehat{Var}(\bar{Y})}} \sim t_{n-1}$ distribution
- Therefore, in practice we replace the $z_{\frac{\alpha}{2}}$ percentile with a $t_{n-1, \frac{\alpha}{2}}$ percentile, so the CI becomes
 - $\bar{y} \pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\frac{N-n}{N} \frac{s}{\sqrt{n}}}$

2.4 Confidence Intervals for μ

- For large samples, $t_{n-1, 1-\frac{\alpha}{2}} \approx Z_{1-\frac{\alpha}{2}}$
- For smaller samples, $t_{n-1, 1-\frac{\alpha}{2}} > Z_{1-\frac{\alpha}{2}}$, so this t distribution CI will be larger
- Using these formulae for large populations correctly depends on two things
 - n
 - How “normal” the population is

2.4 Confidence Intervals for μ

- Some populations are quite normal, e.g. heights of statistics student
- Some populations are very skewed, e.g. income of statistics students 20 years after graduation
- So before you use the CLT (and those formulae for CIs), think about the normality of the population being studied
 - For the time being, it suffices to use the t distribution CIs. Later on, we will see what to do if we are seriously concerned about our normality assumption.

2.4 Confidence Intervals for μ

- We want to estimate the average amount of money students at a particular high school spend on junk food each day
 - There are 2010 students in the high school
 - We sample 100 by SRS
 - The total of observations in the sample is \$2907
 - The total of squared observations in the sample is 154593\$²
 - Find a 95% C.I. for μ
 - What happens if you treat the student population as infinite?

2.5 Sample sizes for μ

- We will say that our goal is to have a C.I. of a certain width, $2d$.
 - What sample size do I need to achieve this width?
- If $Var(\bar{Y})$ is known, the width of the $100(1 - \alpha)\%$ C.I. is
 - $2\sqrt{\frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)} Z_{1-\frac{\alpha}{2}}$
- Hence the appropriate sample size is the n which solves
 - $\sqrt{\frac{\sigma^2}{n} \left(\frac{N-n}{N-1}\right)} Z_{1-\frac{\alpha}{2}} = d$
- Some calculation shows
 - $n = \frac{1}{\frac{1}{N} + \frac{d^2}{\sigma^2 Z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)}$

2.5 Sample sizes for μ

- The problem with this is... we usually don't know σ . How can we estimate it?
 - **Estimate it:** use the sample variance from your pilot survey (assuming you did one...)
 - **Copy it:** use some **historical data (existing studies)** that exists on the topic and plug-in values found there
 - **Guess it:** if you have a reasonable hypothesis for the distribution of the data, that can give you some information regarding the variance.
 - E.g. you believe the population is normal and you have an idea for the range of the data. Then you could estimate σ by $\frac{Range}{4}$ or $\frac{Range}{6}$ because 95% and 99.7% of values from a normal distribution are within 2 and 3 standard deviations of the mean respectively.

2.5 Sample sizes for μ

- Say pre-sample estimate for σ is v
- Then we may use this formula to find an appropriate sample size for us

- $$n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 z^2 \left(1 - \frac{\alpha}{2}\right)} \left(1 - \frac{1}{N}\right)}$$

2.5 Sample sizes for μ

- Hang on... isn't that cheating? What about the t -distribution?
 - Remember, we are finding a sensible value for n **before** we perform the survey
 - So just as we might not know σ , we cannot know what value s will take
 - The best we can do is plug-in an estimate from historical data, your pilot survey or one based on the range of values
- But you may feel that using the Normal distribution to find n is too aggressive
 - you would prefer to play safe and consider the t -distribution

2.5 Sample sizes for μ

- Then to find the appropriate sample size we need to solve
 - $\sqrt{\left(1 - \frac{n}{N}\right) \frac{v^2}{n}} t_{n-1, 1-\frac{\alpha}{2}} = d$ for n
 - i.e. to solve $n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t_{n-1, 1-\frac{\alpha}{2}}^2}}$
- The difficulty here is that n appears on both sides of the equation
 - So **how can we solve this?**

2.5 Sample sizes for μ

- Use trial and error
- Or use Excel/R...
 - Let $N = 1000, v = 40, d = 10, \alpha = 0.05$
 - This table is taken from an Excel spreadsheet
 - We choose the first n which is larger than the right hand side of the equation
 - Thus, we choose $n = 61$

n	$t_{n-1, 1-\frac{\alpha}{2}}$	$\frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t_{n-1, 1-\frac{\alpha}{2}}^2}}$
2	12.70620474	720.9169057
3	4.30265273	228.5171704
4	3.182446305	139.4499289
$\vdots$	$\vdots$	$\vdots$
60	2.000995378	60.20665759
61	2.000297822	60.16721346
62	1.999623585	60.12909785

2.6 Estimating τ

- The sweat and tears we have given working out results for estimating μ will now pay off
- Because τ is related to μ in such a simple way, we simply adapt all the formulae we developed for $\hat{\mu}$
- Note $\tau = \sum_{j=1}^N u_j = N \frac{1}{N} \sum_{j=1}^N u_j = N\mu$
- So we choose our estimator $\hat{\tau}$ to be $N\bar{Y}$

2.6 Estimating τ

- Is $\hat{\tau}$ unbiased?
 - $E(\hat{\tau}) = E(N\bar{Y}) = NE(\bar{Y}) = N\mu = \tau$. Good.
- What is the variance of $\hat{\tau}$?
 - $Var(\hat{\tau}) = Var(N\bar{Y}) = N^2 Var(\bar{Y}) = \frac{N-n}{N-1} \frac{N^2}{n} \sigma^2$
- How can we estimate the variance of $\hat{\tau}$?
 - Use $N^2 \widehat{Var}(\bar{Y}) = N^2 \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n} := \widehat{Var}(\hat{\tau})$

2.6 Estimating τ

- How can we calculate confidence intervals for τ ? **Just keep track of the N^2 ...**
 - If σ is known, we have the following $100 \times (1 - \alpha)\%$ C.I. for τ :
$$\left(\hat{\tau} \pm \sqrt{N^2 \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)} Z_{1-\frac{\alpha}{2}} \right)$$
 - If σ is unknown, a $100 \times (1 - \alpha)\%$ C.I. for τ is given by
$$\left(\hat{\tau} \pm \sqrt{N^2 \left(1 - \frac{n}{N} \right) \frac{s^2}{n}} t_{n-1, 1-\frac{\alpha}{2}} \right)$$

2.6 Estimating τ

- How can we determine an appropriate sample size for estimating τ given τ 's C.I. of a certain width $2d$?
- Say, we have some number v as our pre-sample estimate for σ . We may solve

$$- n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 v^2 z^2 \left(1 - \frac{\alpha}{2}\right)}}$$

$$- \text{Or } n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 v^2 t^2 \left(1 - \frac{\alpha}{2}\right)}}$$

2.6 Estimating τ



- The Statistics Department is concerned that their students spend too much time watching the attractive football played by Tottenham Hotspur F.C.
- They would like to estimate the total time spent each week by students watching Tottenham
- The department has 750 students, from which a sample of 50 is taken. The average amount of time is 10.31 hours and $s^2 = 2.25$
- Estimate the total number of student-hours given to watching the Tottenham and give a 99% C.I.



2.7 Estimating p

- Recall p is the population proportion. This is just a special case of estimating the population mean.
- Each element in the population **either has the characteristic** (e.g. approves of the CEO's job performance; supports Tottenham Hotspur F.C., wears glasses, etc.) **or doesn't**
 - Let element j have value y_j , where $y_j = 1$ if element j has the characteristic, and $y_j = 0$ if not
- Then $p = \frac{1}{N} \sum_{j=1}^N y_j$. That is, **p is a special μ**

2.7 Estimating p

- Thus a good, unbiased estimator for p is
 - $\hat{p} = \frac{1}{n} \sum_{i=1}^n Y_i$, where $Y_i = 1$ if element i has the characteristic and $Y_i = 0$ otherwise
- To find $Var(\hat{p})$, use $Var(\bar{Y}) = \frac{N-n}{N-1} \frac{1}{n} \sigma^2$
 - Since each Y_i is a Bernoulli random variable with probability of success p , in this case $\sigma^2 = p(1 - p)$ (exercise!)
 - $\Rightarrow Var(\hat{p}) = \frac{N-n}{N-1} \frac{1}{n} p(1 - p)$
 - Note this contains the parameter we want to estimate...

2.7 Estimating p

- To estimate $Var(\hat{p})$,
 - consider $\widehat{Var}(\bar{Y}) = \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}^2}{n}$
 - when the Y_i are Bernoulli random variables, $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 = \frac{n}{n-1} \hat{p} (1 - \hat{p})$ (exercise!)
 - $\Rightarrow \widehat{Var}(\hat{p}) = \left(1 - \frac{n}{N}\right) \frac{1}{n-1} \hat{p}(1 - \hat{p})$ is an unbiased estimator for $Var(\hat{p})$

2.7 Estimating p

- To calculate a $100(1 - \alpha)\%$ confidence interval for p , assuming n is large enough, we must use
 - $\left(\hat{p} \pm \sqrt{\left(1 - \frac{n}{N}\right) \frac{1}{n-1} \hat{p}(1 - \hat{p})} t_{n-1, 1-\frac{\alpha}{2}} \right)$
 - Because $\sigma^2 = p(1 - p)$ will be unknown. If it is known we would know p
- What is large enough, in this case?
 - We require $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$

2.7 Estimating p

- Adapting the formulae for finding sample size n , we have

- $$n = \frac{1}{\frac{1}{N} + \frac{d^2}{p(1-p)z_{1-\frac{\alpha}{2}}^2} \left(1 - \frac{1}{N}\right)}$$

- Obviously, this formula contains p , which we don't know. What can we plug in?
 - An “educated guess” from historical data
 - For a conservative sample size, use $p = 0.5$. ($p(1 - p)$ is maximized at $p = 0.5$)

2.7 Estimating p

- We want to estimate the proportion of statistics student affected a disease called gooneritis
 - There are 1500 students
 - We know that about 30% of maths students suffer from gooneritis
 - We plan to estimate p using SRS.
 - What sample size do I need so that the total length of a 95% C.I. is less than 0.02?
 - What should I do if I think statistics students are very different to maths students?

2.8 Using SRS in Practice

- SRS is our first and simplest survey design. It may not be the best choice for your project
 - Are you sure you want a survey? Perhaps an **experiment** would be better?
 - What if you **do not have a frame**?
 - What if SRS is **too expensive/time-consuming** to perform?
 - What if you know the population you want to study has **certain characteristics** that could help you design a more effective sampling scheme?

2.8 Using SRS in Practice

- On the other hand, SRS is a good choice if
 - You have **very little information** about the population other than a frame
 - The person paying you to do the survey **insists you use SRS**. For example, in court cases SRS is often the method of choice because it is “simple” enough for jurors to understand

STAT3003: Survey Methods

3. Stratified Random Sampling

Contents

- 3.1 Introduction
 - Example, definition, motivation
- 3.2 Estimating τ , μ and p
 - Notation, $\hat{\tau}_{st}$, $Var(\hat{\tau}_{st})$, $\widehat{Var}(\hat{\tau}_{st})$, CIs, Satterthwaite's approximation, sample size given allocation, example, estimating μ , estimating p
- 3.3 Sample Allocation
 - Factors, proportional allocation, Neyman allocation, Lagrange multipliers, Optimal allocation, (cautionary) example

3.1 Introduction

- In the previous lecture we covered our first survey design: **simple random sampling**
- We discussed
 - How to **perform it**
 - What the **best estimators** for population mean, total and proportion are, under this design
 - **Properties** of these estimators (bias, variance etc.)
 - How to **determine sample size**

3.1 Introduction

- We will address the same issues this questions this lecture, but for our second sample survey design...
- Stratified Random Sampling

3.1 Introduction

- You wish to estimate the average wage of workers in a city
- The city has a population of 9, with three companies (A, B and C) employing 3 people each
- The breakdown of monthly wages by company is
 - Company A: \$100,000, \$105,000, \$97,5000
 - Company B: \$50,000, \$47,800, \$49,250
 - Company C: \$12,000, \$11,500, \$10,900

3.1 Introduction

- You have two sampling methods available:
 - A **simple random sample**: randomly select 3 people from the population of 9, take the average
 - A **stratified random sample**: randomly select 1 person from each company, take the average
- Which do you expect will produce the closer estimate?

3.1 Introduction

- Why this difference?
- Because the **variance within** each Company is small, but the **variance between** them is large
- Roughly speaking, we shall see that the variance of the estimator under stratified random sampling **depends only on the variance within each stratum**
- Whereas the variance of the estimator under SRS **depends on the population variance**, which includes the variance within strata and variance between strata

3.1 Introduction

- Definition
 - A stratified random sample is one obtained by **partitioning** the population elements into groups called *strata*, then **performing SRS within** each *stratum*
- Thus performing a stratified random sample consists of two parts
 1. Choosing **non-overlapping** strata for your population
 2. Performing multiple, **independent** SRSs

3.1 Introduction

- What's the point?
 - **Reduction of estimator variance**: an estimator under stratified random sampling can have smaller variance (and therefore tighter confidence intervals)
 - **Reduction of cost**: the cost per observation may be reduced if the strata are convenient
 - **Comparison among sub-groups**: the simple random samples among the strata are independent, enabling comparison of estimates
 - **Prevent really bad (unrepresentative) samples**: sampling from strata ensures elements from those strata will be represented in the sample

3.1 Introduction

- Choosing the strata can be difficult and counter-productive – see the “Cautionary Example” at the end of this lecture
- The criteria need to be **specific** so the **strata do not overlap**
- A **poor choice** of strata can lead to the variance of the estimator being larger than that of a simple random sample
- Once the strata have been chosen, you need to **decide on the sample size in each**. There are many ways of doing this (we shall see later)

3.1 Introduction

- It is very important that the simple random samples conducted within each stratum are **independent**
- So **different random sampling schemes** (lots, random numbers etc.) should be used for each stratum
- So that observations in one stratum do not depend on those in another
- This allows later **comparison** of estimates between the strata

3.1 Introduction

- Example
 - You want to conduct a survey to find out how many hours a week the households in your local area watch TV.
 - Your local area contains a public housing estate, a collection of village houses and a new development aimed at young professionals
 - Why might you consider using stratified sampling for your survey?

3.1 Introduction

- Example
 - You wish to conduct a study to investigate the total of worker-hours lost every year due to accidents among all CUHK staff
 - CUHK staff composes of professors, lab technicians, labourers, administrators, monkey-catchers etc.
 - Why might you consider using stratified random sampling for your survey?

3.2 Estimating τ , μ and p

- Before we crack on with defining the estimators and finding their properties, we should define some notation
 - L = Number of strata
 - N_i = Number of sampling units with stratum i
 - N = Number of sampling units in population, so $N = N_1 + N_2 + \dots + N_L$
 - $\bar{Y}_{st}$ = Estimator of population mean from stratified sampling
 - A subscript of i will mean that the parameter/estimator concerned is from stratum i . E.g. $\bar{Y}_i, \mu_i, s_i^2, \tau_i$ etc.

3.2 Estimating τ , μ and p

- This time we start by considering the estimation of the population total, τ
- We have seen previously that $N_i \bar{Y}_i$ is an unbiased estimator of population total τ_i
- Clearly $\tau = \tau_1 + \tau_2 + \cdots + \tau_L$ so $\sum_{i=1}^L N_i \bar{Y}_i$ will be our estimator for τ
- $E\left(\sum_{i=1}^L N_i \bar{Y}_i\right) = \sum_{i=1}^L E(N_i \bar{Y}_i) = \sum_{i=1}^L \tau_i = \tau$, so our estimator is unbiased.
- Let us denote $\sum_{i=1}^L N_i \bar{Y}_i$ by $\hat{\tau}_{st}$

3.2 Estimating τ , μ and p

- How about the variance?
 - $Var(\hat{\tau}_{st}) = Var(\sum_{i=1}^L N_i \bar{Y}_i) = \sum_{i=1}^L Var(N_i \bar{Y}_i)$
because the $\bar{Y}_i$ are independent
 - $\sum_{i=1}^L Var(N_i \bar{Y}_i) = \sum_{i=1}^L N_i^2 Var(\bar{Y}_i)$
 - $= \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$ using the formula from SRS
- Notice that the variance of the estimator depends on σ_i^2 's and not σ^2

3.2 Estimating τ , μ and p

- Again, the variance of our estimator contains parameters we usually do not know, the σ_i^2 's
- Thus we need an unbiased estimator for the variance of our estimator
- Luckily, from our work on simple random sampling, we know an unbiased estimator for

$$\sigma_i^2: \left(1 - \frac{1}{N_i}\right) \hat{\sigma}_i^2$$

3.2 Estimating τ , μ and p

- Plugging our unbiased estimator for σ_i^2 into our formula for $Var(\hat{\tau}_{st})$, we arrive at an unbiased estimator for $Var(\hat{\tau}_{st})$:

- $\widehat{Var}(\hat{\tau}_{st}) := \sum_{i=1}^L N_i^2 \frac{1}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right) \left(1 - \frac{1}{N_i} \right) \hat{\sigma}_i^2$
- $= \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i} \right) \frac{\hat{\sigma}_i^2}{n_i}$

3.2 Estimating τ , μ and p

- Now we know $\widehat{Var}(\hat{\tau}_{st})$, we can create confidence intervals for τ
- If all the stratum sample size are **small**, an approximate $100(1 - \alpha)\%$ C.I. for τ
 - $\hat{\tau}_{st} \pm t_{df, 1 - \frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau}_{st})}$
- The question is, what should we use for the df ?
 - Intuitively, we might guess $df = df_1 + df_2 + \dots + df_L$ where $df_k = n_k - 1$, the df from the SRS in stratum k
 - But this is correct only **if the variances of all the strata are the same**

3.2 Estimating τ , μ and p

- Say we have a linear combination of sample variances, χ'
 - $\chi' = \sum_{i=1}^n k_i \hat{\sigma}_i^2$
- We don't know the distribution of χ' . However, we can approximate it well by a chi-squared distribution whose d.f. is given by
 - $df_{\chi'} \approx \frac{(\sum_{i=1}^n k_i \hat{\sigma}_i^2)^2}{\sum_{i=1}^n \frac{(k_i \hat{\sigma}_i^2)^2}{n_i - 1}}$, where $\hat{\sigma}_i^2$ has sample size n_i
 - This is called **Satterthwaite's Approximation**

3.2 Estimating τ , μ and p

- Thus, in our case,
 - $df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}}$ where $k_h = \frac{N_h(N_h - n_h)}{n_h}$
 - Calculating df will produce a non-integer. Be conservative and round it down
 - Thompson (2012) advocates a rule of thumb:
 - If the strata sample sizes are at least 30, we can avoid calculating df by using the normal distribution
 - $\hat{\tau}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau}_{st})}$

3.2 Estimating τ , μ and p

- Finally, we want to find an appropriate sample size $n = n_1 + n_2 + \dots + n_L$
 - given our desired width of C.I. $2d$
 - Or given our desired bound on the error of estimation, d
- As before, this requires us solving
 - $z_{1-\frac{\alpha}{2}}\sqrt{Var(\hat{t}_{st})} = d$ for n
- But we need to know
 - The variances in each strata, i.e. σ_i
 - the relationship between each n_i and n
- All we know is that n_i is some fraction of n
 - So let us write $n_i = na_i$ for $i = 1, \dots, L$

3.2 Estimating τ , μ and p

- Solving for n , we reach

$$\bullet \quad n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{\sigma_i^2}{N_i - 1}}{\frac{d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{N_i - 1}}$$

- As before, we usually don't know the σ_i s, but we may have reasonable guesses v_i s
 - So plug in the v_i s in place of the σ_i s
- Think about trying to find n if we replaced $z_{1-\frac{\alpha}{2}}$ with $t_{df, 1-\frac{\alpha}{2}} \dots$
 - Can we do it?

3.2 Estimating τ , μ and p

- A politician is thinking of running a TV advert attacking her opponent in a small town.
- She wishes to estimate the total hours of TV watched in the town.
- The town consists of parts A, B and C
 - Part A surrounds a factory. Most households are families with parents working in the factory.
 - Most households in Part B are retired couples.
 - Part C is rural, with most households belonging to farming families

3.2 Estimating τ , μ and p

- The politician's team of statisticians decide to use stratified random sampling to estimate τ , the total hours each household will watch TV each week
- The summary statistics are

Part	N_i	n_i	$\bar{y}_i$	s_i
A	155	20	33.90	5.95
B	62	8	25.12	15.25
C	93	12	19.00	9.36

- Find a 95% C.I. for τ

3.2 Estimating τ , μ and p

- We now move on to estimating the population mean, μ
- Given that $\tau = N\mu$, we can find the properties of our estimator for μ quite easily
- We define our estimator for μ , denoted $\bar{Y}_{st}$, by remembering the fact that $\mu = \frac{\tau}{N}$
 - $\bar{Y}_{st} := \frac{1}{N} \hat{\tau}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \bar{Y}_i$

3.2 Estimating τ , μ and p

- Finding the variance of $\bar{Y}_{st}$ is easy:
 - $Var(\bar{Y}_{st}) = Var\left(\frac{1}{N} \hat{t}_{st}\right) = \frac{1}{N^2} Var(\hat{t}_{st})$
 - So $Var(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1}\right)$
- And so is finding an unbiased estimator for $Var(\bar{Y}_{st})$
 - We replace σ_i^2 's by their unbiased estimators:
 - $\widehat{Var}(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i}\right) \frac{\hat{\sigma}_i^2}{n_i}$

3.2 Estimating τ , μ and p

- The C.I.s for μ quickly follow
- If the sample size in each strata is bigger than 30, we may use

- $\bar{y}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})}$

- If any of the sample sizes are small, we use

- $\bar{y}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})}$

- With $df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}}$ where $k_h = \frac{N_h(N_h - n_h)}{\textcolor{red}{N}^2 n_h}$

3.2 Estimating τ , μ and p

- To find an appropriate sample size n to achieve our given C.I. width d , we again write the sample size in each stratum as a fraction of n : $n_i := a_i n$
- Then we solve $z_{1-\frac{\alpha}{2}} \sqrt{\text{Var}(\bar{Y}_{st})} = d$ for n , which yields

$$n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{\sigma_i^2}{N_i - 1}}{\frac{N^2 d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{N_i - 1}}$$

3.2 Estimating τ , μ and p

- As before, we may treat estimating p as estimating the total number of people in a population who have a particular characteristic
- Given a simple random sample, we saw that $N_i\hat{p}_i$ was a good estimator of N_ip_i the number of people in a population of size N_i who have a particular characteristic.
- Thus, in stratified random sample,
 $N_1\hat{p}_1 + N_2\hat{p}_2 + \cdots + N_L\hat{p}_L$ would be a good estimator of the number of people in a population of size N who have a particular characteristic

3.2 Estimating τ , μ and p

- Thus we define $\hat{p}_{st}$ as our estimator of the population proportion p , where
 - $\hat{p}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \hat{p}_i$
- The variance of $\hat{p}_{st}$ is easy to find, remembering the variance of $\hat{p}_i$ under SRS:
 - $Var(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 Var(\hat{p}_i)$
 - $= \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{N_i - n_i}{N_i - 1} \frac{1}{n_i} p_i (1 - p_i)$

3.2 Estimating τ , μ and p

- Like before, $Var(\hat{p}_{st})$ contains parameters we do not know. We need an unbiased estimator for it
- $Var(\hat{p}_{st})$ is a linear sum of population variances for each stratum i , σ_i^2 , where the population consists of N_i zeros and ones
- In such populations, $\sigma_i^2 = p_i(1 - p_i)$
- Recall that an unbiased estimator for σ_i^2 is $\left(1 - \frac{1}{N_i}\right) \hat{\sigma}_i^2$

3.2 Estimating τ , μ and p

- We have also shown that, when the population consists of zeros and ones, the sample variance can be calculated by $\hat{\sigma}_i^2 = \frac{n_i}{n_i-1} \hat{p}_i(1 - \hat{p}_i)$
- So by replacing $p_i(1 - p_i)$ in our formula for $Var(\hat{p}_{st})$ by $\left(1 - \frac{1}{N_i}\right) \frac{n_i}{n_i-1} \hat{p}_i(1 - \hat{p}_i)$ we find an unbiased estimator for $Var(\hat{p}_{st})$:
 - $\widehat{Var}(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \left(1 - \frac{n_i}{N_i}\right) \frac{1}{n_i-1} \hat{p}_i(1 - \hat{p}_i)$

3.2 Estimating τ , μ and p

- The C.I.s for p quickly follow
- If the sample size in each strata is bigger than 30, we may use

- $\hat{p}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{p}_{st})}$

- If any of the sample sizes are small, we use

- $\hat{p}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{p}_{st})}$

- With $df \approx \frac{(\sum_{h=1}^L k_h s_h^2)^2}{\sum_{h=1}^L \frac{(k_h s_h^2)^2}{n_h - 1}}$ where $k_h = \frac{N_h(N_h - n_h)}{\textcolor{red}{N}^2 n_h}$

3.2 Estimating τ , μ and p

- Finally, we want to find an appropriate sample size $n = n_1 + n_2 + \cdots + n_L$
 - given our desired width of C.I. $2d$
 - Or given our desired bound on the error of estimation, d
- As before, this requires us solving
 - $z_{1-\frac{\alpha}{2}}\sqrt{Var(\hat{p}_{st})} = d$ for n
- But we need to know
 - The true proportion in each stratum, p_i
 - Our chosen allocation, e.g. the a_i s.t. $n_i = na_i$

3.2 Estimating τ , μ and p

- Solving $d = z_{1-\frac{\alpha}{2}} \sqrt{Var(\hat{p})}$ for n yields
 - $$n = \frac{\sum_{i=1}^L N_i^3 \frac{1}{a_i} \frac{p_i(1-p_i)}{N_i-1}}{\frac{N^2 d^2}{z_{1-\frac{\alpha}{2}}^2} + \sum_{i=1}^L N_i^2 \frac{p_i(1-p_i)}{N_i-1}}$$
- Of course, we don't know the p_i s – that's the purpose of the survey after all, but we may have reasonable guesses v_i s
 - So plug in the v_i s in place of the p_i s
- Alternatively, if we have no idea what p_i is likely to be, plug in $p_i = 0.5$

3.2 Estimating τ , μ and p

- This complicated formula rapidly reduces to something simple if we make some reasonable assumptions
- Assume $\frac{N_i}{N_{i-1}} \approx 1$
 - $$n = \frac{\sum_{i=1}^L N_i^2 \frac{1}{a_i} p_i(1-p_i)}{\frac{N^2 d^2}{z^2 \frac{1-\alpha}{1-\frac{\alpha}{2}}} + \sum_{i=1}^L N_i p_i(1-p_i)}$$
- Take $p_i = 0.5$
 - $$n = \frac{\sum_{i=1}^L N_i^2 \frac{1}{a_i}}{\frac{4N^2 d^2}{z^2 \frac{1-\alpha}{1-\frac{\alpha}{2}}} + N}$$
- Use proportional allocation
 - $$n = \frac{N^2}{\frac{4N^2 d^2}{z^2 \frac{1-\alpha}{1-\frac{\alpha}{2}}} + N}$$
- Use a 95% C.I.
 - $$n = \frac{N}{1.04 d^2 N + 1}$$
- Say N is very large
 - $$n = \frac{1}{1.04 d^2}$$

3.3 Sample Allocation

- How should we decide these a_i (equivalently, the n_i , the sample size in the i th stratum)?
- The n_i also appear in the variance of our estimator:
 - $$Var(\hat{t}_{st}) = \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$
 - $$Var(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$
- Our objective is the following: to obtain a specified amount of information at a minimum cost

3.3 Sample Allocation

- Thus the best allocation scheme will be affected by
 - ?
 - ?
 - ?

3.3 Sample Allocation

- Thus the best allocation scheme $a_1, \dots, a_L$ or $n_1, \dots, n_L$ will be affected by
 - The “population” size of each stratum (a sample of 10 from a population of 20 contains more information than a sample of 10 out of 2000000)
 - The variability of observations within the stratum (the higher the variability (σ_i), the larger the sample needed to obtain a good estimate)
 - The cost of obtaining observations from each stratum (the cost may vary from stratum to stratum)

3.3 Sample Allocation

- Choose sample sizes $n_1, \dots, n_L$ that are in the same proportion to the total sample size n as the $N_1, \dots, N_L$ are to the total population size N
- Therefore $n_1 = \frac{nN_1}{N}, n_2 = \frac{nN_2}{N}, \dots, n_L = \frac{nN_L}{N}$
 - And $a_1 = \frac{N_1}{N}, a_2 = \frac{N_2}{N}, \dots, a_L = \frac{N_L}{N}$
- This is simplistic – it does not consider the variability within each stratum
 - If we know about the variances inside each stratum, we can allocate our sample better

3.3 Sample Allocation

- Assume the following
 - You have **no overhead cost**
 - The **cost of collecting data is the same** in each stratum
 - Your **total sample size is fixed at n**
- Then how should we choose the a_i (or n_i)?
- We choose the a_i that **minimize the variance of our estimator**, $Var(\bar{Y}_{st})$

3.3 Sample Allocation

- Mathematically, this means
 - minimizing the function $\frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$
 - over all $n_1, n_2, \dots, n_L$
 - with the constraint $n_1 + n_2 + \dots + n_L = n$
 - This is called Neyman allocation
- This is an example of optimization with constraints
 - The technique to solve such problems uses Lagrange Multipliers

3.3 Sample Allocation

- Consider the problem
 - Maximize/minimize $f(x_1, \dots, x_n)$
 - Subject to the constraint $g(x_1, \dots, x_n) = 0$
- We solve this by
 1. Writing down the **Lagrangian**
$$L(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) - \lambda g(x_1, \dots, x_n)$$
 2. Find the **stationary points** of $L(x_1, \dots, x_n, \lambda)$, i.e. where $\frac{\partial L}{\partial x_i} = 0$ for $i = 1, \dots, n$ and $\frac{\partial L}{\partial \lambda} = 0$
 3. Evaluate $f(x_1, \dots, x_n)$ at each stationary point, **determine which is the maximum/minimum**

3.3 Sample Allocation

- Example
 - Find the maximum and minimum of
$$f(x, y, z) = x + y + z$$
 - subject to the constraint $x^2 + y^2 + z^2 = 1$
- Solution
 - Write down the Lagrangian:
 - $L(x, y, z, \lambda) = x + y + z - \lambda(x^2 + y^2 + z^2 - 1)$
 - Find the stationary points. First find the partial derivatives:
 - $\frac{\partial L}{\partial x} = 1 - 2\lambda x, \frac{\partial L}{\partial y} = 1 - 2\lambda y, \frac{\partial L}{\partial z} = 1 - 2\lambda z,$
$$\frac{\partial L}{\partial \lambda} = -(x^2 + y^2 + z^2 - 1)$$

3.3 Sample Allocation

– Set them to zero and create a system of equations

- $\lambda x = \frac{1}{2}$
- $\lambda y = \frac{1}{2}$
- $\lambda z = \frac{1}{2}$
- $x^2 + y^2 + z^2 = 1$

– Now solve them

- $\Rightarrow \lambda^2(x^2 + y^2 + z^2) = \lambda^2 = \frac{3}{4}$
- $\Rightarrow$ if $\lambda = -\frac{\sqrt{3}}{2}$, $(x, y, z) = \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$
- $\Rightarrow$ if $\lambda = \frac{\sqrt{3}}{2}$, $(x, y, z) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

3.3 Sample Allocation

– Now evaluate $f(x, y, z)$ at these points

- $f\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) = -\sqrt{3}$

- $f\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \sqrt{3}$

– Conclude

- The constrained maximum value of $\sqrt{3}$ occurs when

$$x = y = z = \frac{1}{\sqrt{3}}$$

- The constrained minimum value of $-\sqrt{3}$ occurs when

$$x = y = z = -\frac{1}{\sqrt{3}}$$

3.3 Sample Allocation

- Back to our problem

- minimize the function

$$f(n_1, \dots, n_L) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right)$$

- over all $n_1, n_2, \dots, n_L$

- with the constraint

$$g(n_1, \dots, n_L) = n - \sum_{i=1}^L n_i = 0$$

- Let's solve it

- Write down the Lagrangian

- $L(n_1, \dots, n_L, \lambda) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \frac{\sigma_i^2}{n_i} \left(\frac{N_i - n_i}{N_i - 1} \right) - \lambda(n - \sum_{i=1}^L n_i)$

3.3 Sample Allocation

– Find the partial derivatives

- $\frac{\partial L}{\partial n_i} = -\frac{N_i^3}{N_i-1} \frac{1}{N^2} \frac{\sigma_i^2}{n_i^2} + \lambda$ for $i = 1, \dots, L$ (exercise!)
- $\frac{\partial L}{\partial \lambda} = (\sum_{i=1}^L n_i - n)$

– Set them to zero and create a system of equations

- $\lambda n_i^2 = \frac{N_i^3}{N_i-1} \frac{1}{N^2} \sigma_i^2 \Rightarrow n_i = \frac{1}{\sqrt{\lambda}} \sqrt{\frac{N_i}{N_i-1} \frac{N_i}{N}} \sigma_i$ for $i = 1, \dots, L$
- $\sum_{i=1}^L n_i = n \Rightarrow \frac{1}{\sqrt{\lambda}} = \frac{n}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i-1} \frac{N_i}{N}} \sigma_i}$

– And solve

- $n_i = n \frac{\sqrt{\frac{N_i}{N_i-1}} N_i \sigma_i}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i-1}} N_i \sigma_i}$
- This is the solution of Neyman allocation.
- Trust me, these n_i minimize $Var(\bar{Y}_{st})$

3.3 Sample Allocation

- Neyman allocation is an example of optimal allocation
 - We can optimise over other factors as well
- Assume the following
 - You have a **fixed cost, c_0**
 - The cost of obtaining a single observation from stratum i is **c_i** , $i = 1, \dots, L$
 - Your total budget is **c**
- Then you choose the a_i that minimize $V(\bar{Y}_{st})$ subject to the constraint that
 - **$c_0 + c_1n_1 + c_2n_2 + \dots + c_Ln_L = c$**

3.3 Sample Allocation

- The solution to this (and a question in Problem Sheet 2) is

- $$n_i = \frac{(c - c_0) \left(\sqrt{\frac{N_i}{N_i - 1}} \frac{N_i \sigma_i}{\sqrt{c_i}} \right)}{\sum_{i=1}^L \sqrt{\frac{N_i}{N_i - 1}} N_i \sigma_i \sqrt{c_i}}$$

- As the cost of sampling in a stratum increases, the optimal sample size in that stratum decreases
- If c_i are the same for $i = 1, \dots, L$, this becomes the same solution as Neyman allocation (exercise)

3.3 Sample Allocation

- **A Cautionary Example** : Is it always true that stratified random sampling produces an estimator with smaller variance than SRS?
 - A food distributor wishes to add a new product to the restaurants he supplies
 - So he adds his new product to a sample of restaurants in Hong Kong
 - He supplies 4 major chains: Maxim ($N_1 = 24$ stores), Café de Coral ($N_2 = 36$), Fairwood ($N_3 = 30$) and Yoshinoya ($N_4 = 30$)
 - He wants to estimate μ , the mean sales of the new product per store per month.
 - The distributor only has enough resources to obtain data from $n = 20$ stores

3.3 Sample Allocation

- He decided to use stratified sampling, with each chain acting as a stratum
- As he has no idea what the σ_i^2 are, and the cost of sampling is the same in each stratum, he uses proportional allocation
 - He samples $n_1 = \frac{24}{120} \times 20 = 4$ restaurants from Maxim, $n_2 = \frac{36}{120} \times 20 = 6$ restaurants from Café de Coral, etc.

3.3 Sample Allocation

- He observes the following data of sales in the month
 - Work out $\bar{y}_{st}$, his estimate for the average sales for the month and provide a 95% C.I. for μ

Maxim	Café de Coral	Fairwood	Yoshinoya
94	91	108	92
90	99	96	110
102	93	100	94
110	105	93	91
	111	93	113
	101		
$\hat{y}_1 = 99,$ $s_1^2 = 78.67$	$\hat{y}_1 = 100,$ $s_1^2 = 55.60$	$\hat{y}_1 = 98,$ $s_1^2 = 39.50$	$\hat{y}_1 = 100,$ $s_1^2 = 112.50$

3.3 Sample Allocation

- Now imagine that the distributor had used SRS instead, with exactly the same sample of 20 restaurants (the same data)
 - Work out $\bar{y}$, and calculate the 95% C.I.
 - Compare your answers with those from the stratified sample – what do you notice? What do you think has happened? What can be done to correct this?

3.3 Sample Allocation

- The lesson is
 - Stratifying will not necessarily result in a smaller variance
 - If your reason for stratifying is to have a smaller error of estimation, choose strata whose observations within are likely to be homogeneous
- You may have other reasons for stratifying
 - Convenience
 - You want to estimate parameters for subgroups of the population